SE Lab-5

PES2UG24CS465

BEFORE:

```
Be ready to explain the bugs you found and how you fixed them. You may be called on to present.
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation> python blood_donation.py
Stock: {'A+': 0, 'A-': 0, 'B+': 0, 'B-': 0, 'O+': 2, 'O-': 1, 'AB+': 0, 'AB-': 0}
Units for an A+ patient: 3
Units for a Z+ patient: 0
Is O- universal? True
Low stock groups: ['A+', 'A-', 'B+', 'B-', 'O-', 'AB+', 'AB-']
Donors eligible today: []
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation>
```

1)Pylint report

pylint_report.txt

```
1  ************* Module blood_donation
2  blood_donation.py:1:0: C0114: Missing module docstring (missing-module-docstring)
3  blood_donation.py:17:0: C0116: Missing function or method docstring (missing-function-docstring)
4  blood_donation.py:17:0: C0103: Function name "registerDonor" doesn't conform to snake_case naming style (invalid-name)
5  blood_donation.py:17:18: C0103: Argument name "donorName" doesn't conform to snake_case naming style (invalid-name)
6  blood_donation.py:17:29: C0103: Argument name "bloodGroup" doesn't conform to snake_case naming style (invalid-name)
7  blood_donation.py:17:46: C0103: Argument name "lastDonation" doesn't conform to snake_case naming style (invalid-name)
8  blood_donation.py:17:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
9  blood_donation.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
10 blood_donation.py:27:0: C0103: Function name "canDonate" doesn't conform to snake_case naming style (invalid-name)
11 blood_donation.py:33:0: C0116: Missing function or method docstring (missing-function-docstring)
12 blood_donation.py:33:0: C0103: Function name "recordDonation" doesn't conform to snake_case naming style (invalid-name)
13 blood_donation.py:33:19: C0103: Argument name "donorName" doesn't conform to snake_case naming style (invalid-name)
14 blood_donation.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)
15 blood_donation.py:42:0: C0103: Function name "unitsFor" doesn't conform to snake_case naming style (invalid-name)
16 blood_donation.py:42:13: C0103: Argument name "patientGroup" doesn't conform to snake_case naming style (invalid-name)
17 blood_donation.py:45:11: W0718: Catching too general exception Exception (broad-exception-caught)
18 blood_donation.py:49:0: C0116: Missing function or method docstring (missing-function-docstring)
19 blood_donation.py:49:0: C0103: Function name "isUniversal" doesn't conform to snake_case naming style (invalid-name)
20 blood_donation.py:49:16: C0103: Argument name "bloodGroup" doesn't conform to snake_case naming style (invalid-name)
21 blood_donation.py:50:11: R1714: Consider merging these comparisons with 'in' by using 'bloodGroup in ('O-', 'AB+')'. Use a set instead if elements are hashable.
22 blood_donation.py:53:0: C0116: Missing function or method docstring (missing-function-docstring)
23 blood_donation.py:53:0: C0103: Function name "lowStock" doesn't conform to snake_case naming style (invalid-name)
24 blood_donation.py:58:0: C0116: Missing function or method docstring (missing-function-docstring)
25 blood_donation.py:58:0: C0103: Function name "donorSheetPath" doesn't conform to snake_case naming style (invalid-name)
26 blood_donation.py:63:0: C0116: Missing function or method docstring (missing-function-docstring)
27 blood_donation.py:63:0: C0103: Function name "notifyDonors" doesn't conform to snake_case naming style (invalid-name)
28 blood_donation.py:63:17: C0103: Argument name "phoneList" doesn't conform to snake_case naming style (invalid-name)
29 blood_donation.py:68:0: C0116: Missing function or method docstring (missing-function-docstring)
30
31 ------------------------------------------------------------------
32 Your code has been rated at 4.81/10 (previous run: 4.81/10, +0.00)
33
```

2)bandit report

bandit_report.txt

```
python : [main]  INFO  profile include tests: None
At line:1 char:1
+ python -m bandit -r blood_donation.py > bandit_report.txt 2>&1
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]  INFO  profile include tests: None:String) [], RemoteExcepti
   on
    + FullyQualifiedErrorId : NativeCommandError

[main]  INFO  profile exclude tests: None
[main]  INFO  cli include tests: None
[main]  INFO  cli exclude tests: None
[main]  INFO  running on Python 3.13.2
Run started:2026-10-14 07:00:43.607522+00:00

Test results:
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'bloodbank-demo'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\blood_donation.py:5:19
4
5  OFFICER_PASSWORD = "bloodbank-demo"
6  donors = []

--------------------------------------------------
>> Issue: [B101:assert_used] Use of assert detected. The enclosed code will be removed when compiling to optimised b
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b101_assert_used.html
   Location: .\blood_donation.py:18:4
17 def registerDonor(donorName, bloodGroup, age, lastDonation=None, notes=[]):
18     assert bloodGroup in stock, "Unknown blood group"
19     if age < 18 or age > 65:
```

```
--------------------------------------------------
>> Issue: [B306:blacklist] Use of insecure and deprecated function (mktemp).
   Severity: Medium   Confidence: High
   CWE: CWE-377 (https://cwe.mitre.org/data/definitions/377.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b306-mktemp-q
   Location: .\blood_donation.py:60:11
59     # Helper kept for the lab: not called by the demo
60     return tempfile.mktemp(suffix=".csv")
61

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\blood_donation.py:65:4
64     # Helper kept for the lab: not called by the demo
65     os.system("send-sms " + " ".join(phoneList))
66

--------------------------------------------------

Code scanned:
    Total lines of code: 61
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 2
        Medium: 1
        High: 1
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 3
Files skipped (0):
```

3)Flake8 report

flake8_report.txt

```
1  blood_donation.py:7:80: E501 line too long (82 > 79 characters)
2  blood_donation.py:58:1: E303 too many blank lines (3)
3  blood_donation.py:80:43: E261 at least two spaces before inline comment
4  blood_donation.py:81:13: E225 missing whitespace around operator
5
```

AFTER:

1)pylint report

```
>>
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation> Get-Content lab-5_reports\pylint_report_after.txt
************* Module blood_donation
blood_donation.py:109:0: C0304: Final newline missing (missing-final-newline)

-------------------------------------------------------------------
Your code has been rated at 9.82/10 (previous run: 4.81/10, +5.01)

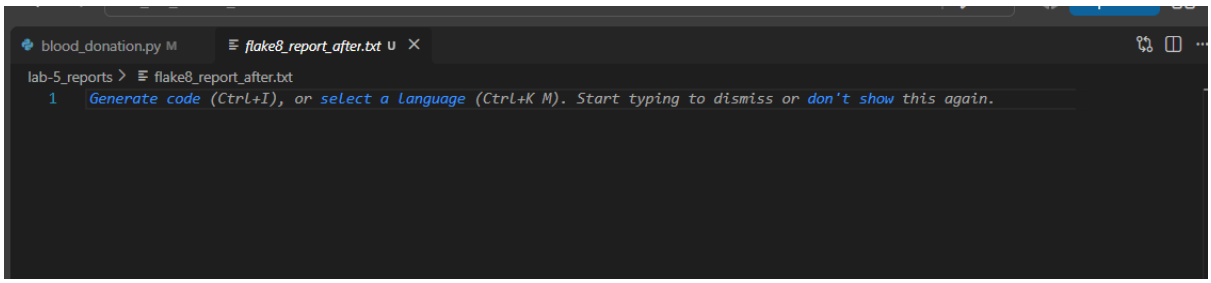
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation>
```

2)Bandit report

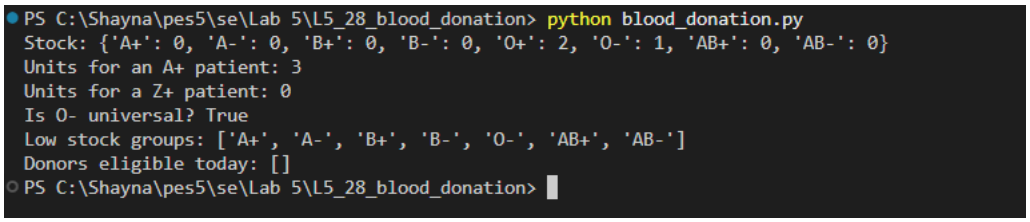
```
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation> Get-Content lab-5_reports\bandit_report_after.txt
        Total lines of code: 84
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
PS C:\Shayna\pes5\se\Lab 5\L5_28_blood_donation>
```

3)flake8



Running:



Issues table:

Issue	Type	Lines	Description	Fix Approach
Mutable default argument	Bug	14	notes=[] is shared across function calls.	Changed default to None and initialized inside the function.
Broad exception	Code Quality	49	except Exception catches unnecessary exceptions.	Replaced with except KeyError.
Hardcoded password	Security	6	Password was stored directly in the source code.	Removed the hardcoded password.
Use of assert	Security	79	assert was used for input validation.	Replaced with an explicit condition and ValueError.
Insecure temporary file	Security	90	tempfile.mktemp() can create a security risk.	Replaced with tempfile.NamedTemporaryFile().
Unsafe command execution	Security	96	os.system() executes a shell command.	Removed the unused command-execution helper.
Long line	Style	7	A line exceeded the allowed length.	Split the dictionary across multiple lines.
Incorrect whitespace	Style	80-81	Missing whitespace around an operator and before a comment.	Corrected spacing.
Missing final newline	Style	108	The file did not end with a newline.	Added a final newline.